Diabetes Technology Expert Program

DBLG2

1. Introduction

Welcome to this video on the DBLG2 System.

DBLG2 is an automated insulin delivery algorithm developed in France that offers a unique approach to diabetes management. It is currently available in selected European countries and may expand to additional regions in the future.

The DBLG2 System consists of three components:

- the Dexcom G7 sensor
- the Kaleido pump
- and the DBLG2 algorithm on an Android phone.

In the near future, a connection to the Dana-i insulin pump will also be available.

A key difference compared to the previous DBLG1 version is where the algorithm operates. With DBLG1, the algorithm was housed on a dedicated handset. With DBLG2, the algorithm runs directly on an Android phone.

At the time this course was created, the app was not yet available for iPhone. Availability for iOS will follow in the near future.

In the general videos on automated insulin delivery or AID systems, you have already learned valuable information such as comparing AID systems using the CARES reference framework, choosing an AID system, reading reports, and managing various situations like hypo and hyperglycemia, high-fat meals, exercise, illness, and travel.

In the upcoming videos, we will delve into the specifics of the DBLG2 System, providing you with a detailed understanding of its operation and step-by-step instructions. This comprehensive education will equip you with the knowledge needed to navigate the system effectively.

Please be aware that the content presented here may change without prior notification. The User Manual for the medical device remains the primary document for ensuring the safe operation, and it's important to note that these videos are not intended as medical advice. Always consult your healthcare provider before making any adjustments to your therapy.

The videos have been created by Dr. Inge van Boxelaer, the founder of Diabetotech and an endocrinologist at the St-Lucas hospital in Ghent, Belgium. The content has been reviewed by Dr. Sarah Siegelaar, an endocrinologist at Amsterdam UMC in the Netherlands. Their expertise and impartiality ensure that you receive honest opinions about the pros and cons of the DBLG2 System.

By the end of these videos, you will have a clear understanding of:

- How DBLG2 performs according to the CARES paradigm
- How the DBLG2 System works in practice
- How to generate and interpret reports in the YourLoops platform
- And how to manage specific situations like hypo and hyperglycemia, high-fat meals, sports, illness, alcohol, and travel when using DBLG2

As a healthcare provider, this knowledge will enable you to guide DBLG2 users more effectively, and as a DBLG2 user, you will be better prepared for a successful start while gaining useful tips and tricks to simplify your life with the DBLG2 system. Good luck!

2. DBLG2 System according to the CARES Paradigm

Welcome to this video where we will explore the DBLG2 System based on the CARES paradigm.

We will discuss how the DBLG1 algorithm calculates and adjusts the automated insulin delivery, when it reverts back to manual mode, specific education tips and sensor issues, and how the system shares its data. After that we will go over the official indications and software updates of the DBLG2 system.

2.1 How the DBLG1 system works based on the CARES paradigm

1.1 Calculate

The DBLG2 System operates on a unique calculation algorithm that takes into account various factors to determine insulin delivery. Here are the key aspects of the calculation process:

- The system delivers microboluses of insulin every 5 minutes based on current and predicted glucose levels, and the total daily insulin dose to maintain glucose levels as close as possible to the defined target glucose. It also takes into account announced physical activity and meals.
- During initialization, half of the pre-set total daily dose of insulin is allocated to basal insulin, while the other half is allocated to meal insulin.
- The target glucose range can be adjusted between 100 and 130 mg/dL or 5.6 and 7.2 mmol/l.
- Autocorrection boluses are administered every 5 minutes, when the glucose level is predicted to exceed 180 mg/dL or 10 mmol/L. If the glucose level is predicted to fall below the hypoglycemic threshold, basal insulin is paused, and rescue carbohydrates are recommended.
- The system incorporates both short-term and long-term learning algorithms. The short-term self-learning algorithm analyzes trends over the past 2-3 days, while the long-term algorithm considers trends over the past 3-6 weeks.

The DBLG2 algorithm also features a unique meal management algorithm, which involves recommendations for administering the meal bolus. Here are the key aspects of the meal management algorithm:

- In some cases, the system divides the meal bolus into two administrations based on sensor glucose levels and the meal type.
- In the case of normal glucose levels, the first part of the bolus is given immediately, and the second part is administered after 30 minutes. For high-fat meals, the second part is administered after 60 minutes. The post meal management algorithm might additionally deliver correction micro-boluses between the first and second parts of the meal bolus. In this case, the second part of the bolus is subsequently modified to take into account the active insulin.
- If hypoglycemia is predicted, the second part of the bolus can be delayed up to 90 minutes or can even be canceled.

- If hyperglycemia is detected at the start of the meal, the system can give the bolus as a single administration. Additional autocorrection boluses may be given if necessary.
- After a meal, basal insulin can be replaced by autocorrection boluses, which typically last for a maximum of 3 hours.

1.2 Adjust

With the DBLG2 algorithm, you have the ability to adjust various parameters to personalize your insulin delivery. Here are the parameters you can modify:

1. The target glucose level. The default value is set at 110 mg/dl or 6.1 mmol/l, but you can adjust it between 100 and 130 mg/dl or 5.6 and 7.2 mmol/l.
2. The aggressiveness at meals determines the amount of insulin delivered at mealtimes. The default value is 100%, and you can set it between 50% and 200%. Adjusting this parameter allows you to have more or less insulin for meals, similar to adjusting the carbohydrate ratio in other automated insulin delivery systems.
3. The aggressiveness in hyperglycemia controls the response to high glucose levels. The default value is 100%, and it can be adjusted between 43% and 186%. Increasing or decreasing this parameter affects the insulin doses of the autocorrection boluses.
4. The aggressiveness in normoglycemia modulates basal insulin delivery during normal glucose levels. The default value is 100%, and you can adjust it between 59% and 147%. Modifying this parameter makes the modulation of basal insulin more or less aggressive.
5. The hypoglycemia threshold determines the glucose level at which the system recommends rescue carbohydrates and adjusts basal insulin delivery. The default value is 70 mg/dl or 3.9 mmol/l, but it can be set between 60 and 85 mg/dl or 3.3 and 4.7 mmol/l.
6. The Total Daily Dose of insulin is the main setting to fine tune the overall glucose management by the system. It should be reduced in case of too frequent rescue carbs recommendations and hypoglycemia, or on the contrary, it may be increased in case of too frequent hyperglycemia (postprandial or fasting). Note that while you can modify the total daily dose of insulin between 8 and 90 units per day, frequent adjustments may affect the auto learning of the system, so it is recommended to avoid frequent changes, and make adjustments in steps of no more than 10%.

In addition to the parameters mentioned earlier, the DBLG2 algorithm offers additional modes to accommodate specific situations:

- Physical Activity Mode is designed for physical exercise, such as sports, intense gardening, house work etcetera. A physical activity can be declared for the current day or the following day. When enabled, the target glucose level, as well as the hypoglycemia threshold, are adjusted to better suit the demands of the activities. The target value and thresholds are increased by 70 mg/dl or 3.9 mmol/l. If necessary, earlier rescue carbohydrates will be recommended. The duration of this mode is adjustable within a 0 to 24 hour range. After the activity, the DBLG1 algorithm will slowly lower the target value back to the normal target glucose level. The type and intensity of the physical activity impacts the degree of aggressiveness reduction around physical activity.
- Zen Mode can be activated if you want to avoid hypoglycemia, and be bothered as little as possible with the need to eat rescue carbohydrates. During Zen mode, the target value can be increased by 10-40 mg/dl or 0.6-2.2 mmol/l. The duration of Zen mode can be adjusted within a 1-8 hour range via the settings.

When starting with the DBLG2 system, you need to enter your total insulin dose, weight, and the average number of grams of carbohydrates you consume per standard meal. You may enter three standard meals for breakfast, lunch, and dinner. These parameters can be edited later if needed.

However, there are certain parameters that the algorithm calculates and cannot be adjusted.

- These include the basal insulin rate, carbohydrate ratio, insulin sensitivity and duration of insulin action.
- During the set-up, you are required to enter a basal safety profile, but this is only used in manual mode or open-loop.
- The hyperglycemia threshold is fixed at 180 mg/dl or 10 mmol/l. In the DBLG2 app you are able to change the hyperglycemia threshold, but this does not impact the algorithm. It will only affect when the glucose value on your DBLG2 app is displayed in orange.

1.3 Revert

If the connection to the insulin pump or glucose sensor is lost for more than 30 minutes, including during the sensor warmup period, the DBLG2 system will return to manual mode. You can check the connection status in the system status screen, and if it shows "signal loss” or “waiting pump”, it means the connection is not established.

While theoretically, the connection should be maintained within a 2-meter range or 6 feet, it is recommended to keep your handset as close to the sensor and **insulin** pump as possible for reliable communication.

In manual mode, there is no suspend-for-low function available. The basal insulin delivered in manual mode is referred to as the basal safety profile, and you have the ability to adjust it directly from your handset. Additionally, if your handset is still connected to the pump, you can use it to give an insulin bolus or set a temporary basal rate.

Once the connection to the pump or sensor is reestablished, the system will automatically resume auto mode or closed-loop mode.

1.4 Educate

When starting an automated insulin delivery system like DBLG2, there are general education points applicable to all automated insulin delivery systems, as well as ~~**specific**~~ education points specific to the DBLG2 system. Let's go over them:

For general education in automated insulin delivery systems, please refer to the generic module. These include the following key points:

- Use fewer carbohydrates to correct hypoglycemia.
- Consider an infusion set problem if blood glucose levels remain high for an extended period without an apparent reason. "If in doubt, change it out."
- Trust the automated insulin delivery system and let it do its work.
- Give a meal bolus approximately 15 minutes before eating.
- Activate the Activity mode 1 to 2 hours prior to engaging in exercise.
- Temporarily stop insulin delivery if you disconnect the pump for more than 15 minutes.
- Respond to system alarms promptly.

- Disable automatic updates on your mobile phone to avoid disruptions.
- Seek peer support for additional guidance and insights.

Now let's focus on the specific education points for the DBLG2 system:

- During initialization, half of the total daily dose of insulin is allocated to basal insulin, while the other half is allocated to meal insulin. If this allocation doesn't align well with your actual insulin needs, it's recommended to adjust the total daily dose of insulin accordingly. If you typically use a smaller amount of bolus insulin compared to basal insulin (e.g., 30/70% if you follow a low-carb diet), increasing the total daily dose of insulin by 5-10% at initialization can be beneficial. On the other hand, if you typically use a larger amount of bolus insulin compared to basal insulin (e.g., 70/30% if you consume a higher amount of carbohydrates), reducing the total daily dose of insulin by 5-10% at initialization can be considered.
- It’s advised to enter your meals at least 15 minutes before the meal, although the actual insulin bolus will only be given 6 minutes before the meal. Don’t forget to validate the insulin bolus at that time, or the insulin will not be given.
- Keep your phone close to your body or within arms reach, and charge it overnight.
- Have rescue carbohydrates on hand and be familiar with the amount of carbohydrates they contain (e.g., 5, 10, or 15 grams) for quick and easy administration.
- Adjusting the total daily insulin dose is the most effective parameter for fine-tuning the DBLG1 algorithm. However, avoid making frequent adjustments as it resets the autolearning.
- You can enter carbohydrate information either through the bolus calculator or as rescue carbs. If you choose the latter option, the system will not provide additional insulin for the carbohydrates consumed.
- You can enter past and future meals and physical activities. Even if you forgot to enter a meal or activity, it's still beneficial to input it later for autolearning by the algorithm.
- The DBLG2 system offers two additional modes: Zen mode, which reduces the risk of hypoglycemia, and Confidential mode, which temporarily prevents data transmission to YourLoops.

- After a meal and during autocorrect boluses, the basal insulin level decreases, as per the unique behavior of the DBLG2 algorithm.
- In the YourLoops reports, the total daily insulin dose is divided into a percentage for basal insulin and a percentage for bolus insulin. The autocorrection boluses are added to the bolus percentage, which means a relatively higher bolus percentage is frequently observed in the reports.

By familiarizing yourself with these education points, you will have a better understanding of how to optimize the DBLG2 system and manage your diabetes effectively.

1.5 Sensor

The DBLG2 system utilizes the Dexcom G7 sensor for glucose monitoring. If your sensor does not show a number or arrow, or your readings do not match your symptoms, use your finger prick meter to make diabetes treatment decisions. When in doubt, get your meter out.

If there is a significant discrepancy between the sensor glucose reading and the finger prick measurement, you may consider calibrating your Dexcom sensor. However, it is important to calibrate only when your glucose levels have been stable and within the target range for the past few minutes. You do not have to calibrate, but you can.

When you start with a new sensor, you will not receive any Dexcom readings or alarm/alerts until the 2-hour or 30-minute warmup period has finished. During this time, use your finger prick meter to make treatment decisions.

1.6 Share

With the DBLG2 system, you have the option to share insulin and glucose data with your healthcare provider and family members via YourLoops. Real-time monitoring of glucose values is possible for both parties.

Data is automatically uploaded to YourLoops, a web-based data visualization platform, via mobile data (4G) or Wi-Fi.

Users can choose Wi-Fi as the preferred method for data transfer. When this option is selected, data is only sent to YourLoops while the device is connected to a Wi-Fi network. This reduces mobile data usage and allows transmission through a more stable connection. However, data updates may be delayed when no Wi-Fi is available. If no Wi-Fi connection has been detected for two consecutive weeks, technical and YourLoops data will automatically be transmitted via the available mobile network.

If there is no internet connection — for example during a flight — data will not be transmitted. However, the system continues to function normally, as it only requires Bluetooth to connect to the pump and sensor.

Healthcare providers and family members can access YourLoops through their respective login credentials.

The DBLG2 system provides a unique feature called Confidential Mode, allowing users to opt out of sharing their data for a specific period of 3 hours, 1 day or 3 days. It's important to note that when Confidential Mode is activated, the data will not be recoverable in YourLoops afterwards.

2.2 Indications

The DBLG2 algorithm is CE-labeled for use in adults with type 1 diabetes, and FDA-approved for individuals aged 12 years and older with type 1 diabetes. It requires a prescription from a healthcare professional and is intended for single-patient use only.

The system is suited for individuals with a total daily insulin requirement between 8 and 90 units. It is compatible with rapid-acting insulin analogues NovoRapid and Humalog at a concentration of 100 U/mL.

Please note that the DBLG2 system is not approved for use during pregnancy or in individuals undergoing dialysis.

2.3 Software updates

When a new version of the DBLG2 application becomes available, a message is displayed in the app. Simply select “Update” to download and install the latest version.

After the application has been updated, it is important to perform a few checks. First, confirm that the sensor and the pump have automatically reconnected and are functioning correctly. Then verify whether Loop mode is active. If it has been switched off during the update process, make sure to reactivate it.

Regarding operating system updates, do not upgrade your phone to an unsupported version. Always wait until Diabeloop has confirmed that the new operating system version is compatible with the DBLG2 system.

If your phone installs a new operating system version, take the following steps: restart the DBLG2 application, grant any newly requested permissions, confirm that the sensor and pump have reconnected properly, and ensure that Loop mode is active. If necessary, switch it on again.

In conclusion, the DBLG2 system integrates various components to provide automated insulin delivery. With its advanced algorithm and adjustable parameters, the DBLG2 system provides flexibility to meet individual needs. If you have any questions or concerns, you can seek assistance from your healthcare provider, the dedicated Facebook group or contact the distributor's customer service. Remember, you are not alone in your diabetes management journey, and there are resources available to support you every step of the way.

3. Linking components together

In this video, we will demonstrate the process of establishing connections between the components of the DBLG2 System.

- This system functions by continuously measuring glucose levels with the Dexcom sensor every 5 minutes.
- The glucose data is then transmitted to the DBLG2 handset, which calculates and sends insulin instructions to the Kaleido insulin pump.
- Additionally, the DBLG2 app communicates with the YourLoops server, where therapy settings and therapy data is stored.

During this video, we will cover the steps to

- link the Dexcom G7 sensor to the DBLG2 app,

- establish a connection between the Kaleido insulin pump and the DBLG2 app,
- and guide you through the process of creating a YourLoops account

These connections are crucial for the seamless functioning of the DBLG2 System.

3.1 Linking the Dexcom G7 sensor to the DBLG2 app

Before you begin, keep in mind that when the Dexcom G7 sensor is used with the DBLG2 app, it cannot be used with the Dexcom app at the same time.

To pair your Dexcom G7 sensor, you must first sign in to your Dexcom account. Ensure that your phone is connected to the internet for this purpose.

From the Main Menu, go to System, and tap Pair. Then, select Dexcom Account. Enter your Dexcom account credentials to sign in. If you don't have an account yet, tap Create Account and follow the prompts to register.

Next, tap Unbox and Enter Number. Once you have removed the sensor from its packaging, tap Next. Enter the sensor number printed on the applicator, then tap Enter. If you prefer, you can tap Scan Code to scan the QR code instead, and then follow the on-screen instructions. Tap Insert and Start.

Now it's time to insert your Dexcom G7 sensor: Gather the required materials. Then, follow the instructions for sensor placement.

After inserting the sensor, return to your smartphone and tap Connect. The app will begin pairing with the sensor. This process can take up to 30 minutes, so keep your smartphone nearby throughout the connection period. During this time, the Dexcom G7 will not provide glucose readings, alarms, or alerts. You can tap I Understand to close the message while the connection continues in the background. Be sure to watch for the pairing authorization notification and approve it when it appears. On the System screen, the sensor status will change to Searching while the connection is being established.

When pairing is complete, the sensor will automatically begin its warm-up period, which lasts 30 minutes. A countdown timer will be displayed on both the Home and System screens. Glucose readings, alarms, and alerts will not be available while the sensor is warming up. You can use this waiting period to set up your insulin pump.

Once the warm-up is finished, the sensor will begin transmitting glucose readings to the DBLG2 app, and all glucose alarms and alerts will become available.

3.2 Linking the insulin pump to the DBLG2 app

At the time this course was created, connection with the Dana-i insulin pump was not yet available. Therefore, we will demonstrate the process using the Kaleido pump.

Although you will receive a Kaleido handset, it's important that you do not pair your Kaleido pumps with the Kaleido handset when using the DBLG2 System. If you were already using the Kaleido pump before starting on the DBLG2 System, you will first need to unpair both of your pumps from the Kaleido Handset before you are able to pair your pumps with the DBLG2 app.

To pair a Kaleido insulin pump with the DBLG2 app, first prepare the pump. Fill a new cartridge and insert it into the pump. Then insert the infusion set, connect the pump to the infusion set, and attach the pump to your body.

Open the DBLG2 app and proceed to the pump pairing section. Depending on your Android version, the app may require access to location services to enable Bluetooth pairing. If needed, the app will request this permission during the process.

When your pump appears on the screen, select it. After the cartridge has been inserted, the pump's color and serial number will be displayed. Carefully verify that both the color and the serial number match your pump before continuing. When the Bluetooth pairing request appears, confirm the pairing.

The application will ask you to confirm that both the cartridge and infusion set are new, and to specify the cannula length. You will then prime the cannula and start insulin delivery.

Once the pump is started, insulin delivery begins according to the programmed basal safety profile. At this stage, Loop mode is not yet active, meaning automatic insulin adjustments are not operating.

In the System Status screen, a green check mark appears next to the paired pump. You will also see the remaining insulin level and the battery level. By selecting "More," additional information such as the pump's serial number becomes visible.

On the home screen, the pump icon will no longer be greyed out, indicating that the pump is successfully connected and active.

After a maximum of 3 days, you'll need to change your infusions set, insulin reservoir and swap over your Kaleido pump. Unlike other patch pumps, a Kaleido pump does not get thrown away, but you'll have 2 pumps that you'll use alternately.

To switch pumps, first disconnect your current pump by selecting **Stop** in the System Status screen. Then disconnect the infusion set, remove the cartridge from the pump, and place the used pump on charge.

Next, insert a new, fully filled cartridge into your second pump, attach a new infusion set, and pair this pump with the DBLG2 app.

Always stop the pump before removing the insulin cartridge. If you remove the cartridge without stopping the pump first, an alarm will be triggered.

Importantly, you do not need to stop Loop mode when switching pumps. The pump can be stopped and replaced, and the cartridge changed, without interrupting Loop mode.

3.3 Creating a YourLoops account

The way to create a YourLoops account as a user or a healthcare provider is different.

- As a user, you should only create a YourLoops account by entering your email address and password when you initialize your DBLG2 app for the first time. You can then use these credentials to log on to www.your-loops.com.

- As a healthcare provider, you can easily create an account at www.your-loops.com. After confirming your email address and logging in as a healthcare provider, your account will display a patient list. This patient list will be empty until some patients share their data with you or with a care team whom you are part of.

To get started as a healthcare provider, you need to create a new team and add your hospital's contact details. YourLoops will automatically assign a unique identification code to your team. You can then invite patients by entering their email addresses and providing them with the identification code.

- As a patient, you can either click on the invitation email to access YourLoops or log in to YourLoops directly. You will be prompted to accept the invitation and enter the 9-digit team code. Once the patient accepts the invitation, their data can be tracked in real time through your patient list.
- The patient list can be sorted by name, first name, time in range, and time below range. However, please note that this data is only available for the last 24 hours and not for the past 14 days, which would be more clinically relevant for healthcare providers.
- When you click on a patient, you will see the same view as the patients themselves, with the patient's name displayed at the top left. Additionally, there are two buttons at the top for patients and treatment teams.

3.4 Linking the DBLG2 app to YourLoops

When you first start up the DBLG2 app, you will be prompted to create a YourLoops account, which will link your information to the YourLoops cloud.

- After creating your account, you will receive an email.
- It's important to remember the email address and password you used for your YourLoops account, as you will need them for logging in or resetting your PIN.

If you want to use the DBLG2 application on a new phone, first open the app on your current device and go to **Profile**, then **My Account**, and select **Manage Account**. Log out of your account and follow the on-screen instructions to stop and unpair the pump. Do not stop the sensor. If you stop the sensor session, you will not be able to reuse the same sensor with the new phone.

Next, install the DBLG2 application on your new phone. Initialize the app using the same activation code, your YourLoops credentials, and the therapy settings available in YourLoops, or complete the setup with the assistance of your healthcare professional.

When changing phones, your historical data remains accessible in YourLoops. However, data stored locally on the previous phone cannot be transferred to the new device.

By correctly linking the sensor, pump, and YourLoops account, you ensure that all components of the DBLG2 system work together as intended — safely, reliably, and efficiently.

Once these connections are in place, the system can continuously measure glucose, calculate insulin needs, and adjust delivery in a coordinated way. Taking the time to set everything up properly gives you a strong foundation for successful automated insulin delivery and allows you to use the full potential of the DBLG2 system with confidence.

4. How the DBLG2 System works

Welcome to this lesson on the functionality of the DBLG2 system. Controlling your insulin delivery is done via the DBLG2 app.

In this video, we will delve into the functionality of the DBLG2 app in general, and then explore how you can use it in both loop mode ON or Auto Mode, and loop mode OFF or Manual Mode.

4.1 DBLG2 app

When you open the DBLG2 app for the first time, you will be guided through an onboarding process. During this setup, you must enter your activation code to enable the system. You will also be asked to provide your height, weight, total daily insulin dose, and the typical carbohydrate amounts for breakfast, lunch, and dinner.

Finally, you must configure a basal safety profile before the app can be fully activated.

Once the sensor and pump have been paired, as explained in the previous lesson, you can activate Loop mode from the System Status screen.

On the home screen, you will see the device status, current glucose value, event history, and insulin delivery information. There is also an event button and four navigation icons at the bottom of the screen: Home, Events, System, and Profile.

Selecting the event button allows you to announce rescue carbohydrates, physical activity, or a meal. Under “More,” you can also record an insulin injection delivered with a pen.

The Home icon returns you to the home screen.
The Events icon opens the events overview screen.
The System icon takes you to the System Status screen.
The Profile icon opens the profile menu.

Within the Profile menu, you can access and adjust account details, treatment settings, view statistics, manage notifications, activate Zen mode or Night mode, and configure data sharing. Additional sections include Help and Contact, Privacy and Security, and About.

Most key configurations are located under Treatment Settings. Here, you can review and modify:

- Medical information such as weight, height, and total daily insulin dose.
- Targets and thresholds, including hypo- and hyperglycemia thresholds and your target glucose level.
- Algorithm aggressiveness, which can be adjusted separately for hyperglycemia, normoglycemia, and per meal.
- Typical meals, where you define carbohydrate amounts for small, medium, and large meals for breakfast, lunch, and dinner.
- The basal safety profile, which is used whenever Loop mode is turned off.

In the Statistics section, you can review your time in range, average total daily insulin dose, and estimated HbA1c over a selected time period.

Under Notifications, you can configure alerts for both Loop mode on and Loop mode off situations, adjust phone notification settings, and test alarm sounds.

- In Loop mode alerts, you can disable notifications for rescue carbohydrate recommendations and rapid glucose rise.
- In Loop off alerts, you can define how long signal loss with the Dexcom sensor is tolerated before receiving an alert, and you can set low and high glucose alert thresholds.
- Through phone notification settings, you can manage general phone notifications and Do Not Disturb mode.

Together, these menu options and settings allow you to personalize and configure your DBLG2 app according to your individual treatment needs and preferences.

4.2 Loop mode ON (Auto Mode or Closed Loop mode)

Starting Loop Mode

In the DBLG2 app, Auto Mode is called Loop mode ON. To activate this on the DBLG2 app, simply tap "Start" in the Loop mode section in the System Status menu. This is only possible if the pump and sensor are properly paired.

Using Physical Activity and Zen mode

To utilize the Physical Activity mode, simply tap the event button on the home screen and choose physical activity. From there, you can input the name or type of the activity, start time, duration, and intensity of your planned physical activity, and then confirm your selections.
In Loop mode ON, you have the option to start or stop Zen mode directly from the home screen. Zen mode will automatically end after 3 hours, but you have the flexibility to adjust the duration, up to a maximum of 8 hours.

Adjusting parameters

During the intialization phase, the DBLG2 app will prompt you to enter your total daily insulin dose, weight, and average grams of carb intake during breakfast, lunch, and dinner, as well as your safety basal profile.

Later you can adjust the following parameters via the treatment settings:

- Your weight and total daily dose
- Your target glucose level and hypoglycemia level
- Your aggressiveness settings
- Your average meal sizes
- And your basal safety profile.

Delivering a bolus

To administer a meal bolus, simply press the Event button, choose Meal and follow these steps:

- First select the time of your meal. Remember, it is recommended to announce meals 15 minutes in advance to allow the algorithm to account for them. If you forget to announce your meal, you can enter a meal in the past. In such cases, the system might suggest an alternative bolus based on your current glucose level.
- Next, enter the meal type and amount of carbs you are going to eat. You can also choose a preset portion size, like small, average or large meal. The proposed portion sizes can be set via the settings.
- Finally you can select if your meal is high fat or not, and press Confirm. DBLG2 will display a “bolus recommendation alert” with a certain amount of insulin, which you can change or confirm.
- While it is possible to override the system's advice, it is generally recommended to follow the recommendations for optimal learning and effectiveness.
- Finally, you will be asked to confirm again the administration of the meal bolus.
- On the homescreen, you'll see "Bolus in progress" during the bolus. If you tap the bolus event on the home screen, you have the option to stop the bolus that is currently in progress, if needed. This can also be done from the Events screen.

Here are some things to keep in mind about giving a meal insulin bolus with the DBLG2 system:

- If a meal is announced in advance, the alert with the recommended bolus will be displayed 6 minutes before the designated mealtime. Remember to validate the alert bolus recommendation, to ensure the bolus is administered.
- The DBLG2 algorithm may decide to deliver the bolus in one or two parts, and if it suggests a two-part bolus, you don't need to confirm the second part separately. It will be automatically delivered.
- If the algorithm is currently giving an autocorrection or a meal bolus, you will not be able to give a bolus. You will need to wait for it to complete, before declaring a meal to administer a meal bolus.
- If you want to enter carbohydrates without receiving insulin, such as in the case of treating hypoglycemia, you have to input them as rescue carbohydrates via the menu.

Suspending insulin

To temporarily stop the pump—for example, when disconnecting for contact sports or using a jacuzzi—follow these steps:

Go to **System Status** and select. On the status screen, you will see that insulin delivery has stopped, and on the home screen the pump icon will appear grey.

When the pump is stopped, Pump Control Mode is disabled. It is not possible to deliver a bolus or adjust the basal rate.

To restart insulin delivery, return to the status screen and select **Start**.

4.3 Loop Mode OFF (Manual Mode or Open Loop mode)

If Loop Mode of the DBLG2 system is not active, the pump can still be controlled manually via the DBLG2 app. In this mode, the integrated bolus calculator is not available.

From the status screen, the **Pump Control** button allows you to start or stop a bolus and to initiate or discontinue a temporary basal rate.

To adjust basal insulin delivery when Loop Mode is OFF, go to the Profile menu, select **Treatment**, then **Basal Safety Profile**. Here, you can review and modify the basal insulin rates that will be used when Loop Mode is OFF.

If you are using the Kaleido pump and do not have your phone with you, only the safety basal profile will be delivered. It will not be possible to administer meal or correction boluses without the phone.

In this video, we explored the functionality of the DBLG2 system, with a focus on the DBLG2 app and its features and settings. Remember, you can do it!

If you ever need additional guidance, don't forget to consult the manual or your healthcare provider for further assistance.

5. Creating and interpreting reports

Welcome to this informative video on creating and interpreting reports using the YourLoops data visualization platform in the DBLG2 system.

By now, you should have already learned how to create your YourLoops account and establish a connection with your DBLG2 handset, which enables data forwarding to YourLoops every 5 minutes.

In this video, we will cover the following topics:
- The various reports available in YourLoops
- And how to effectively interpret these reports

By the end of this video, you will have a comprehensive understanding of the reporting features in YourLoops, allowing you to make the most of this valuable tool. So let's dive in and explore the world of reports of the DBLG2 system!

5.1 Types of reports

Within YourLoops, there are 5 main tabs: Dashboard, Daily, Trends, Profile, and Devices.

The Dashboard provides a centralized view of a patient's diabetes management and automated insulin delivery performance for the past two weeks.

Starting on the left side of the screen, you can review key glucose metrics, specifically Time in Range and Time Below Range. Below these, details about the patient's carbohydrate intake and insulin delivery are presented, covering meal and rescue carbohydrates, total insulin administered, and the distribution of basal, correction, and bolus insulin. The Time in Loop Mode shows the average time spent in automated basal delivery. At the bottom, additional glucose indicators, including average glucose, standard deviation, CV, and GMI, or estimated HbA1c, provide insight into overall glycemic control.

The middle section brings together device information and therapy management details. At the top, the Devices section identifies the patient's connected technologies, namely the CGM, handset, and insulin pump. The Last Updates section provides a history of recent changes to therapy settings and patient parameters. Below this, Sensor Usage highlights CGM utilization, while the Cartridge Changes log records recent pump cartridge replacements.

To the right of these sections, Monitoring Alerts displays information related to time spent outside the target range, hyperglycemia, severe hypoglycemia, and data transmission. Clicking on the cog icon provides a shortcut to the Profile tab, which will be discussed later. The Medical Files section enables clinicians to create and manage patient-related documents.

On the far right of the screen, the Messages feature supports direct communication between the patient and healthcare provider.

The Daily tab presents a detailed view of a patient's glucose management and insulin therapy over a single day.

At the top of the tab, navigation controls allow you to move between dates and select the day you wish to review. A Device Settings button is also available in the upper-right corner. Clicking this button reveals additional details for the displayed day.

The main area of the tab shows the glucose curve for the selected day in relation to the target range, helping clinicians identify periods of hypoglycemia, hyperglycemia, and time spent within range. Events such as the activation of Zen Mode, sports activities, and system alerts are displayed along the timeline and provide additional context for glucose fluctuations throughout the day. Move your cursor over the graph to see the exact glucose value at a specific point in time.

Directly below the glucose graph, the Bolus and Carbohydrates section shows insulin bolus deliveries and carbohydrate entries. Hovering over the color-coded bars reveals additional details, including the exact amounts delivered or entered.

At the bottom, the Basal Rates section provides a visual representation of basal insulin delivery throughout the day. By hovering over the graph, you can view basal rate start and end times, Loop Mode status, and the amounts of insulin delivered.

You can scroll through the glucose graph and Bolus and Carbohydrates section using either the mouse or the scroll bar at the bottom.

On the right side of the screen, a summary panel presents key daily metrics, including Time in Range, Time Below Range, Total Declared Carbohydrates, Total Delivered Insulin, and Time in Loop Mode. Additional glucose statistics, such as average glucose, standard deviation, and CV, are displayed as well.

It is important to note that reviewing data over a longer period often provides more meaningful insights than analyzing a single day. The Trends tab is designed to support this broader view of patient outcomes.

The **Trends tab** offers an Ambulatory Glucose Profile (or AGP). At the top, you can adjust the time range, choosing from options like 1 week, 2 weeks, 4 weeks, 3 months, or filtering specific weekdays. By default, the display shows 100% of the measurements, but you can customize this percentage as desired.

- The 80% data corresponds to the light blue delineations or the 10th-90th percentile, familiar from platforms like Libreview.
- The 50% data represents the dark blue delineations or the 25th-75th percentile, which healthcare providers often focus on to identify trends more easily.

Below the glucose profile, the Rescue Carb Intakes and Manual & Pen Bolus section summarizes the frequency of rescue carbohydrate treatments and manual or pen-delivered boluses over the selected period. The data is grouped by time of day, which makes it easier to identify recurring events and potential opportunities for therapy optimization. A summary of statistics on the right side includes Time in Range, Time Below Range, and average sensor usage.

The Profile tab contains patient information that may support diabetes management and care planning.

On the left side of the tab, three profile sections are available: Information, Range, and Alerts. The Information section is displayed by default. The top area of the presents the patient's name, date of birth, gender, weight, height, email address, equipment start date, HbA1c, glucose units, and insulin type.

Below these patient details is the **Lead Clinicians** section. This area identifies the healthcare providers responsible for managing the patient’s treatment and includes an option to add a clinician. The lower part of the page contains Additional Information fields that can be used to document relevant patient information, such as non-insulin medications, diet, profession, hobbies, physical activity, weekly exercise hours, and open comments. These fields help provide broader context about the patient’s lifestyle and health status.

The **Range** page allows clinicians to define and manage the glucose thresholds used throughout the platform for reporting and analysis. Meanwhile, the **Alerts** page is used to configure monitoring alert thresholds individually for a patient or apply standardized settings defined by the care team.

Finally, the **Devices tab** provides an overview of your handset, pump, and sensor types, as well as your specific settings on the handset. It also tracks changes made to the settings and devices over time.

Beyond these tabs, the **Download Report** button in the top-right corner enables you to select a predefined or custom date range. Once the period is selected, YourLoops generates a report that can be downloaded in either PDF or CSV format. The report includes an overview of Time in Range and other key metrics, cartridge changes, daily charts, and a summary of settings.

By exploring the tabs and utilizing the reporting features in YourLoops, you can gain valuable insights into your data and effectively monitor your diabetes management.

5.2 Interpreting reports

With the DBLG2 system, there are several parameters that can be adjusted to customize the algorithm. It's important to note that the most impactful parameter for the algorithm is the total daily dose of insulin, which can have a significant effect. However, modifying this parameter will reset the auto learning in the long term.

Therefore, if you have been using the loop mode for an extended period and your overall glucose trend is satisfactory, it is rather recommended to consider adjusting other parameters.

The second most influential parameter that can be modified is meal aggressiveness. This parameter allows you to fine-tune how the algorithm responds to meal-related insulin delivery. There is a different aggressiveness factor for breakfast, lunch and dinner. By adjusting a meal aggressiveness factor, you can effectively influence the algorithm's behavior and optimize its response to meals.

Here's a roadmap for interpreting the reports provided by YourLoops, following the guidelines from the general automated insulin delivery module:

1. Assess Glycemic Information: Start by evaluating the Time in Range and Time Below Range over the past 2-4 weeks on the Trends Tab.

- Ensure that treatment goals are being met, with Time In Range above 70% and Time Below Range below 4%.
- Pay attention to the amount of recommended rescue carbohydrates. If a significant number of rescue carbs are recommended without a clear cause,

you can consider adjusting the total daily insulin dose or the low threshold to reduce the frequency of these recommendations. The amount of recommended carbs is based on the user's weight, so modifying this parameter can have an impact on the amount of the recommendations.

- Also, take note of the average amount of carbohydrates entered per day, as this reflects how well the user is accurately entering their meals. However, it's important to note that the YourLoops reports do not provide information on the average number of meal boluses or correction boluses entered in a day.

2. Optimize Automated Insulin Delivery Settings: Analyze the Ambulatory Glucose Profile on the Trends Tab to identify trends of hypo- or hyperglycemia. Adjust the measurement view to focus on the 50% and 80% measurements for clearer trend analysis. Check if these trends are related to boluses on the Daily tab.

- If you notice trends after meal boluses, it's important to assess various factors such as timing of announcing the meal and accurate carbohydrate counting, before adjusting the aggressiveness at meals.
- If you observe trends outside of meals, the most influential parameter is the total daily insulin dose. Before modifying the total daily insulin dose, ensure to verify if the actual insulin dose (visible in the Dashboard tab) differs from the set total daily insulin dose(accessible in the Device tab).
  - For example, here is a case where the set total daily insulin dose is significantly lower than the actual total daily insulin dose issued by the DBLG1 system. The mean Time in Range is only 57%, and despite autocorrection boluses, nocturnal glucose levels remain challenging to manage. Increasing the total daily insulin dose can be a suitable approach for achieving better glycemic control.
  - Likewise, if prolonged hyperglycemia persists despite autocorrection boluses, increasing the total daily insulin dose by 5-10% can be beneficial.
  - On the other hand, if a significant number of rescue carbohydrates are frequently recommended, it may be appropriate to reduce the total daily insulin dose by 5-10%.
  - When adjusting the total daily insulin dose, it's important to reset the aggressiveness back to 100% across all parameters, especially if recent adjustments have been made.

    - Remember, altering the total daily insulin dose resets the auto-learning algorithm, so it's advisable not to make frequent changes. Other parameters that can be adjusted to influence glucose levels outside of meals include the aggressiveness for normoglycemia or hyperglycemia, as well as the target value. Keep in mind that a higher target value not only leads to reduced insulin delivery but also results in more frequent recommendations for rescue carbohydrates.

Note: It's recommended to review the correctness of the total daily insulin dose setting 2 to 3 days after starting the loop mode.

- For instance, if high amounts of rescue carbohydrates are frequently recommended, reducing the total daily dose by 5-10% may be appropriate.
- If the duration of hyperglycemia is prolonged despite multiple microboluses (without experiencing hypoglycemia), increasing the total daily dose by 5-10% may be beneficial.

3. Guide Behavioural Recommendations: To ensure optimal usage of the automated insulin delivery system, it's essential to review the available reports and consider the following behavioural aspects:

- In the Dashboard Tab: Take note of the sensor wear duration and the time spent in Loop mode to ensure adherence and proper system usage.
- In the Daily Tab: Assess the adherence to pre-meal bolusing, addressing hypo- and hyperglycemia, and the appropriate utilization of Physical Activity and Zen Mode.
- Last but not least, ensure that the alarms and reminders are properly configured according to the user's preferences and needs. Please note that these settings are not available in YourLoops and should be checked separately on the DBLG2 app. Via the Profile menu, go to notifications and then select loop mode off alerts.

4. Review Pump Settings and Emergency Plans:

- Verify that the basal safety profile aligns approximately with the basal insulin in Loop mode. The basal safety profile can only be checked on the DBLG2 app, while the current insulin requirements can be found in the Dashboard and Trends tab of the reports.

- Also, verify that the set average number of carbohydrates for different meals is realistic. While it doesn't have to be perfect, it's important to avoid significant deviations. After all, many people stop counting carbs themselves and just enter the recommended carbs per meal.
- Document all pump settings from the Device tab and establish an emergency plan, including the usage of insulin pens, in the event of a pump or handset malfunction. Ensure easy access to insulin pens for such situations.

To conclude: users of the DBLG2 System can generate reports via YourLoops. This provides valuable insights into their glycemic control and system performance. Interpreting these reports and making adjustments to automated insulin delivery parameters can be a trial-and-error process. It is recommended to review the reports after approximately two weeks to evaluate the impact of parameter changes and make any necessary adaptations. This resilience in continuously monitoring and adjusting the system settings ensures optimal performance and improved glycemic management.

6. Managing special situations

Welcome to this video on managing special situations with the DBLG2 System.

In this video, we will provide specific tips and guidelines for individuals using the DBLG2 System in various situations such as hypo- and hyperglycemia, high-fat meals, sports, illness, alcohol and travel. These recommendations supplement the general guidelines for managing automated insulin delivery systems. For more comprehensive information on these topics, please refer to the general module on automated insulin delivery systems.

6.1 Hypoglycemia

In cases of hypoglycemia, the DBLG2 system provides recommendations for consuming rescue carbohydrates and suggests the appropriate amount in grams. It is advisable to maintain a list of your favorite rescue carbs and their corresponding carbohydrate content for quick reference. The recommended amount of rescue carbohydrates is based on your set weight.

If you consume additional rescue carbohydrates beyond what is suggested, you might consider declaring these carbs as a meal.

With an automated insulin delivery system like the DBLG2 system, you will require fewer carbohydrates to raise your blood glucose levels compared to managing hypoglycemia without the system, especially if you have minimal insulin on board. You can check your insulin on board or active insulin on the home screen.

To further prevent future hypoglycemic episodes, you can adjust the system settings to reduce insulin delivery by activating Zen mode or Physical Activity Mode, temporarily setting a higher target value, or decreasing the aggressiveness of insulin delivery.

6.2 Hyperglycemia

In cases of mild hyperglycemia, it is recommended to trust the DBLG2 system and allow it to work autonomously. However, if hyperglycemia persists, it is best to act:

- If you forgot to enter a meal, you can still input the details into the DBLG2 and let the algorithm calculate the required insulin dose. The system allows you to enter meals retroactively.
- If you suspect that you may not have received insulin due to a disconnected infusion set or are unsure about the cause of hyperglycemia, it is advisable to administer a manual correction bolus, using the "Pump control" button via the System Status. Additionally, you can temporarily increase the aggressiveness to address prolonged hyperglycemia.

In the case of persistent and severe hyperglycemia, it is important to consider potential issues with the infusion set. The rule of thumb is "If in doubt, change it out." Furthermore, the DBLG2 system will trigger an alarm if your blood glucose level rises by 60 mg/dL or 3.3 mmol/L within 30 minutes.

If you need to administer an insulin bolus using an insulin pen instead of the system, it is essential to inform the DBLG2 algorithm. To do so, go to the **Event** button, select **More**, and enter the details of the insulin pen bolus.

To ensure preparedness, it is advisable to carry enough supplies for any necessary equipment changes. Having an extra inserter for placing infusion sets is beneficial.

It is recommended to set a reminder in your phone to change the infusion set every 2-3 days. This reminder cannot be set on the DBLG2 app, although the handset will give you a reminder when your insulin cartridge has expired.

6.3 High-fat meals

The DBLG2 algorithm provides a convenient solution for handling high-fat meals. When entering your carbohydrate intake on the handset, you can easily indicate a high-fat meal. In most cases, the DBLG2 algorithm will then split your meal bolus and delay the administration of the second part of the bolus.

In situations where the automatic splitting of the bolus does not work optimally, there are alternative approaches you can consider. One option is to enter only 70% of the total carbohydrates in the meal, allowing the remaining 30% to be accounted for by the system's autocorrection boluses. Another approach is to manually split the bolus by administering 50% of the carbohydrate dose before the meal and the remaining 50% two hours afterward.

These strategies can help address the specific challenges posed by high-fat meals and optimize the management of glucose levels with the DBLG2 system.

6.4 Exercise

When exercising with the DBLG2 system, you can declare the Physical Activity, activate Zen Mode, and or set a higher target glucose value if needed. If you declare a Physical Activity before eating, the DBLG2 algorithm will automatically reduce your meal bolus if needed.

For lowering your basal insulin on board, there are several options:

1. Physical Activity Mode: It is recommended to notify the DBLG1 algorithm well in advance about your planned physical activities by declaring Physical Activity. This allows the system to adjust insulin delivery and prevent excess insulin on board during exercise. If you also enter carbohydrates before exercising, the DBLG2 algorithm will modify the meal bolus accordingly and potentially adjust subsequent meals. If you notify the DBLG2 algorithm about your physical activity late, you may receive an immediate

recommendation for rescue carbohydrates to raise your blood glucose to the higher target value.

2. Zen Mode: If you are unable to announce your physical activity in advance, you can utilize Zen mode to temporarily reduce insulin delivery. This can help to prevent hypoglycemia during exercise.
3. Activity Mode and Zen Mode Combination: In certain situations, you can declare a Physical Activity and turn on Zen mode to further adjust your target glucose range. This can provide additional flexibility in managing your blood sugar levels during exercise.
4. Suspending Insulin Delivery: If needed, you can suspend insulin delivery entirely by stopping and then disconnecting your pump. However, it's crucial to inform the algorithm when you take such actions. It is generally advised not to go without insulin for more than 2 hours to prevent the risk of ketoacidosis.

Keep in mind that these strategies serve as a starting point, and it's important to monitor your glucose trends during exercise. Adjustments to insulin dosages may be necessary based on individual observations and experiences.

6.5 Illness

When you're ill, it is generally recommended to keep using the DBLG2 system and consider increasing the total daily insulin dose.

If you experience prolonged hyperglycemia with elevated ketones, it is necessary to revert to manual mode and contact your healthcare provider.

6.6 Alcohol

When consuming alcohol, you can consider temporarily adjusting your settings to minimize the risk of hypoglycemia, such as increasing your target glucose level, lowering aggressiveness, utilizing Zen mode or declaring a Physical Activity to reduce insulin delivery.

It is worth noting that increasing carbohydrate intake is not effective in preventing hypoglycemia when consuming alcohol. The automated insulin delivery system will compensate for any rise in glucose levels by increasing the administration of insulin.

6.7 Travel

When traveling with the DBLG2 System, it is important to be well-prepared and have an ample supply of necessary items. Here are some key considerations:

- Remember to pack the charger for your phone, Kaleido connection cable, adapter and charging dock and, if needed, a suitable travel plug for the country you are visiting.
- If you're using the Kaleido pump during extended travels, it's also a good idea to take your Kaleido handset with you as a backup device.
- When flying, you can use your phone's airplane mode to turn off your cellular connection. Airplane mode does not deactivate your Bluetooth connection, so your DBLG2 system will keep working. During airplane mode, your data will not be uploaded to Yourloops, but once you turn it off, all data will be uploaded to Yourloops when in Wi-Fi or cellular range.
- If you encounter different time zones during your travel, the DGLG2 app should automatically detect and adjust the time accordingly, if the option "Automatic time zone" is activated in your phone's settings. We advise you to regularly check that the time on your DBLG2 is correct, particularly when you are traveling across multiple time zones.

By keeping these considerations in mind, you can have a smooth travel experience while using the DBLG2 System.

In conclusion, optimizing the DBLG2 system requires proactive management of various situations, and by following the tips provided and seeking support when needed, users can effectively navigate hypo/hyperglycemia, meals, exercise, illness, alcohol consumption, and travel.

You are capable of effectively managing your diabetes with the DBLG2 system. Remember to reach out for support and guidance whenever needed, as you have the resources and strength to navigate this journey successfully.

7. Case report

The DBLG2 System is a relatively new automated insulin delivery system. Despite our inability to find a user review on YouTube at present, we are eager to offer insights from Nina Joachim who shares her personal experience using the DBLG1

System, which is an earlier version of the algorithm on a dedicated handset. From approximately minute 23 onwards, the discussion focuses specifically on the algorithm. Nina explains how the system works in practice and highlights several features she particularly appreciates.

One advantage she mentions is the ability to enter meals after eating if you forget to bolus beforehand—a feature that is not available in most other automated insulin delivery systems. She also discusses how she adjusts the algorithm's aggressiveness setting during menstruation to help manage the temporary increase in insulin requirements. This level of personalization is one of the distinctive features of the DBLG1 and 2 system.

Please click on the link below this lesson to watch the full review and learn how the system performs in everyday life.

Video link: https://www.youtube.com/watch?v=KlchDGuH3OI&t=1392s

Congratulations on completing this course on the DBLG2 system! By equipping yourself with knowledge and understanding, you have taken a significant step towards effectively managing your diabetes. This achievement is important as it empowers you to make informed decisions, optimize your treatment, and enhance your overall well-being. Remember, you are capable of successfully navigating your diabetes journey, and we commend you for your dedication and commitment to your health.